

```
from itertools import permutations

word1 = "SEND"
word2 = "MORE"
result = "MONEY"

letters = set(word1 + word2 + result)

for digits in permutations(range(10), len(letters)):

    mapping = dict(zip(letters, digits))

    # First letters cannot be zero
    if mapping['S'] == 0 or mapping['M'] == 0:
        continue

    def value(word):
        number = 0
        for letter in word:
            number = number * 10 + mapping[letter]
        return number

    num1 = value(word1)
    num2 = value(word2)
    num_result = value(result)

    if num1 + num2 == num_result:
        print("Solution:")
        print("SEND =", num1)
        print("MORE =", num2)
        print("MONEY =", num_result)
        print(mapping)
        break
```

IDLE Shell 3.10.0

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Python 3.10.0 (tags/v3.10.0:b494f59, Oct 4 2021, 19:00:18) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/hi/OneDrive/Desktop/AI/expp.py =====
Solution:
SEND = 9567
MORE = 1085
MONEY = 10652
{'Y': 2, 'N': 6, 'R': 8, 'M': 1, 'D': 7, 'S': 9, 'O': 0, 'E': 5}
>>>

Ln: 10 Col: 0

expp.py - C:/Users/hi/OneDrive/Desktop/AI/expp.py (3.10.0)

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 if num1 + num2 == num_result:
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Ln: 34 Col: 0